# Hughes

October 30, 2026

**Overall Instructions:** Due by Friday, Oct. 30th, 11:00pm EST. Please submit a .pdf of your answers via Canvas, including a url to your .ipynb notebook in a public GitHub repository. Use the provided LATEX template for your submission. Please name your Canvas submission as Last-name.pdf, (e.g. Trujillo.pdf). If you need help with LATEX or GitHub, please visit during office hours or post questions to the Canvas. Include your code in a (well-documented, commented, clear) assignment02.ipynb file. Scores for the code-based questions below will be based on your written responses and code.

**Problem 1 (20 points)** In this question we will work with degree-controlled random graph null models.

(a) The Configuration Model (5 points). First, implement the configuration model in Python. Your function should take a list of integers as a degree sequence, and return a graph where each vertex has the assigned degree. You graph may contain duplicate edges and self-loops.

(b) The Chung Lu Model (5 points). Next, implement the Chung Lu random graph model. Remember that this model creates an edge between nodes i and j with independent probabilities defined by vertex degree and the total number of edges in the graph: pij = kikj 2m , where 2m := i ki. Your implementation should take a list of integers as a degree sequence, and return a simple graph with approximately the desired degree sequence.

(c) Model Differences (5 points). For the degree sequence de- grees=[8,5,5,4,3,2,2,2,1,1,1,1,1,1,1], generate a thousand ran- dom graphs using the Configuration Model and the Chung Lu model, then:

(d) Plot the average adjacency matrix for each model. This should be a heatmap of: A = 1 N N i=1 Ai where N is the number of generated random graphs, and Ai is the adjacency matrix of the ith graph. (Hint: matplotlib's imshow is useful for displaying matrix data.)

(ii) Choose two graph metrics: one that should be approximately the same across graphs gen- erated by both models, and one that should diverge significantly between the two models. For each metric, make a plot (histogram, kernel-density estimate, or cumulative distribution function) showing the distribution of the metric across all generated graphs from each model. Note: For this stage you may remove self-loops and duplicate edges from your configuration model graphs if it helps make a direct comparison to Chung Lu graphs.

(iii) Discussion (5 points). Discuss the above results briefly. Why do the models produce similar results under some metrics and different results under others?

[1]:
```
import networkx as nx
import matplotlib.pyplot as plt
import numpy as np
```

```
from collections import Counter
from itertools import combinations
import logging
import networkx as nx
import matplotlib.pyplot as plt
from scipy.stats import binom
from scipy.special import comb

def configuration_model_from_degree_sequence(degree_sequence,␣
 ↪return_simple=True):
    """
    Generate a random graph using the configuration model from a given degree␣
 ↪sequence
    without using the NetworkX built-in function.

    The configuration model generates a random graph that preserves the degree
    sequence of nodes by assigning "stubs" or "half-edges" to each node and
    randomly pairing these stubs to form edges. This process can result in
    graphs with self-loops and parallel edges, which can be removed if needed.

    Parameters
    ----------
    degree_sequence : list of int
        A list representing the degree of each node in the graph. The sum of
        the degrees in this sequence must be even for the configuration model
        to create a valid graph.

    Returns
    -------
    G : networkx.MultiGraph
        A multigraph generated from the given degree sequence. The graph may
        contain self-loops and parallel edges, which are allowed in the
        configuration model.

    Notes
    -----
    - This method works by assigning "stubs" or "half-edges" to nodes based on
      their degree and then randomly pairing them to form edges. The resulting
      graph can have self-loops and parallel edges.
    - Self-loops and parallel edges can be removed post-generation if a simple
      graph is required using NetworkX's `nx.Graph(G)`.
    - The degree sequence must have an even sum for a valid graph construction.
      If the sum of the degrees is odd, no graph can be constructed.

    Time Complexity
    ---------------
    The time complexity is O(E), where E is the number of edges in the graph.
```

```
    Example
    -------
    >>> degree_sequence = [3, 3, 2, 2, 1, 1]
    >>> G = configuration_model_from_degree_sequence(degree_sequence)
    >>> nx.is_graphical(degree_sequence)
    True
    """

    # Check if the degree sequence is valid (sum of degrees must be even)
    if sum(degree_sequence) % 2 != 0:
        raise ValueError("The sum of the degree sequence must be even.")

    # Create stubs list: node i appears degree_sequence[i] times
    stubs = []
    for node, degree in enumerate(degree_sequence):
        stubs.extend([node] * degree)

    # Shuffle stubs to randomize the pairing process
    np.random.shuffle(stubs)

    # Initialize an empty multigraph
    G = nx.MultiGraph()

    # Add nodes to the graph
    G.add_nodes_from(range(len(degree_sequence)))

    # Pair stubs to create edges
    while stubs:
        u = stubs.pop()
        v = stubs.pop()

        # Add the edge to the graph
        G.add_edge(u, v)

    if return_simple:
        # Convert the multigraph to a simple graph (remove parallel edges)
        G_simple = nx.Graph(G)  # This removes parallel edges by default

        return G_simple

    else:
        return G


def chung_lu_model_from_degree_sequence(degree_sequence, return_simple=True):
    """
```

```
    Generate a random graph using the Chung Lu model from a given degree sequence
    without using the NetworkX built-in function.

    The Chung Lu model generates a random graph that preserves the degree
    sequence of nodes by...

    Parameters
    ----------
    degree_sequence : list of int
        A list representing the degree of each node in the graph. The sum of
        the degrees in this sequence must be even for the Chung Lu model
        to create a valid graph.

    Returns
    -------
    G : networkx.Graph
        A graph generated from the given degree sequence.

    Time Complexity
    ---------------
    The time complexity is O(M+N), where M is the number of edges in the graph and N the number of nodes
    """
    #Instatiate G:
    G = nx.Graph()
    N=len(degree_sequence) #Total number of nodes by taking the length of the degree sequence
    G.add_nodes_from(range(N))

    # Check if the degree sequence is valid (sum of degrees must be even)
    if sum(degree_sequence) % 2 != 0:
        raise ValueError("The sum of the degree sequence must be even.")

    #First calculate the total number of edges in the graph by taking the sum of the degree sequence
    total_edges = sum(degree_sequence)

    #Ensure degree sequence is sorted:
    degree_sequence=np.sort(degree_sequence,descending=True)

    for i in range(0,N-1):#for all nodes
        j=i+1#Look at the next possible node to connect to...
        #and, get the probability of connecting to this node:
        pij = min((degree_sequence[i] * degree_sequence[j]) / (total_edges),1)

        #while there are additional valid other nodes available to connect to...
```

```
        while j < N and pij > 0:
            if (pij < 1):#(i.e., while the probability is not 100% or 0%)
                r = np.random.rand()#draw a random chance
                #and calculate the following to grab one of these additional␣
 ↪other nodes:
                j += int(np.floor(np.log(r)/np.log(1-pij)))

            if j < N:#while there are additional other nodes to process...
                #get the additional other node's probability of connecting to:
                q = min((degree_sequence[i] * degree_sequence[j]) /␣
 ↪(total_edges),1)
                r = np.random.rand()#draw a random chance
                if r < q/pij:#if the random chance is less than the probability␣
 ↪of connecting to this new other node relative to the simple next in the␣
 ↪sequence:
                    G.add_edge(i,j)#add an edge to the new other node
                pij=q#update the probability of connection with the last␣
 ↪processed node
                j+=1#move forward relative to the new other node
    return G

#(b) Testing
degrees=[8,5,5,4,3,2,2,2,1,1,1,1,1,1,1]
chung_lu_graph = chung_lu_model_from_degree_sequence(degrees)
print(chung_lu_graph.degree)
#(c) Testing
chung_lus = 0
configs = 0
num_edges_list_config=[]
num_edges_list_chunglu=[]
cc_list_config=[]
cc_list_chunglu=[]
pos = []
for i in range(1,1001):
    cc_list_config.append(nx.
 ↪average_clustering((configuration_model_from_degree_sequence(degrees))))
    cc_list_chunglu.append(nx.
 ↪average_clustering((chung_lu_model_from_degree_sequence(degrees))))
    num_edges_list_config.
 ↪append((configuration_model_from_degree_sequence(degrees)).number_of_edges())
    num_edges_list_chunglu.
 ↪append((chung_lu_model_from_degree_sequence(degrees)).number_of_edges())
    configs += (nx.
 ↪adjacency_matrix(configuration_model_from_degree_sequence(degrees)))
    chung_lus += (nx.
 ↪adjacency_matrix(chung_lu_model_from_degree_sequence(degrees)))
```

```
    model=configuration_model_from_degree_sequence(degrees)
    pos = nx.spring_layout(model)

configs = configs/1000
chung_lus = chung_lus/1000

print("Depicting the configuration nodel graphically to show any self-loops␣
 ↪that can occur: ")
fig2 = plt.plot()
nx.draw_networkx(model, pos, node_size=150, edgecolors='.2', with_labels=True,␣
 ↪font_size='x-small', edge_color='.6')


fig, ax = plt.subplots(1,2,figsize=(8,3.5),dpi=200)

im1 = ax[0].imshow(configs.todense(),cmap='viridis',vmin = 0, vmax = 1)
ax[0].set_title("Configuration Model Adjacency Matrix Heat Map of 1000␣
 ↪Iterations")
ax[0].grid(True)
ax[0].autoscale(False)

im2 = ax[1].imshow(chung_lus.todense(),cmap='viridis',vmin = 0, vmax = 1)
ax[1].set_title("Chung Lu Model Adjacency Matrix Heat Map of 1000 Iterations")
ax[1].grid(True)
ax[1].autoscale(False)


fig.tight_layout()
fig.colorbar(im1)
fig.colorbar(im2)
plt.show()

# (c) (ii)
#For the metrics, are there any specific ones we want here?
# Could I do just degree distribution? and total of edges?

N=len(degrees)

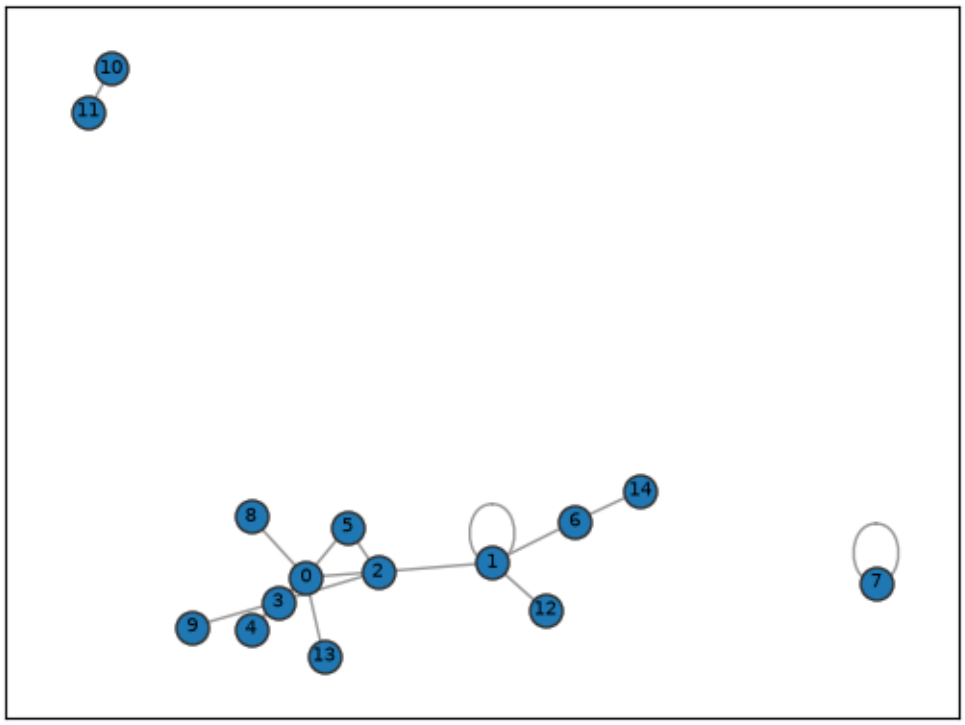
#---Plotting the Total Number of Edges 'L'
#Starting the plotting process
fig, ax = plt.subplots()#Note: fig is the plot container, while ax (i.e., the␣
 ↪'axes') is the plot itself
ax.hist(num_edges_list_config,label="Configuration Model␣
 ↪Sample",density=True)#Plotting the configuration model graph empirical data
ax.hist(num_edges_list_chunglu,label="Chung Lu Model␣
 ↪Sample",density=True)#Plotting the Chung Lu model graph empirical data
```

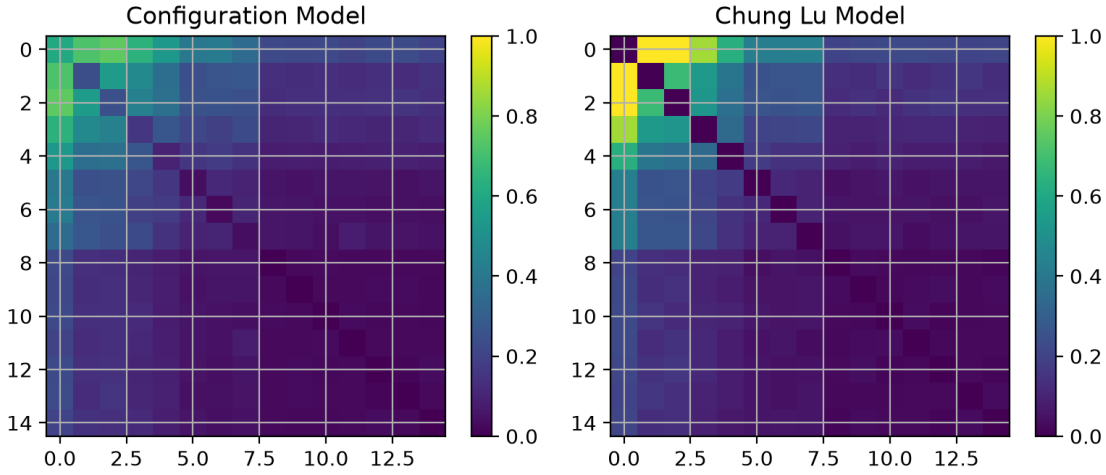
```
#ax.plot(L,P_l, color = 'r',label="Theoretical Distribution") #Plotting the␣
 ↪Theoretical
#Specifying figure details below:
handles, labels = ax.get_legend_handles_labels()
ax.legend(handles=handles, labels=labels,loc="upper right",fontsize="small")
ax.set_title("Configuration Model vs. Chung Lu Model The Total Number of Edges:␣
 ↪'L'")
ax.set_xlabel("L")
ax.set_ylabel("Probability P_L")
true_min=min(min(num_edges_list_config), min(num_edges_list_chunglu))
true_max=max(max(num_edges_list_config), max(num_edges_list_chunglu))
ax.set_xlim(true_min,true_max)
plt.show()

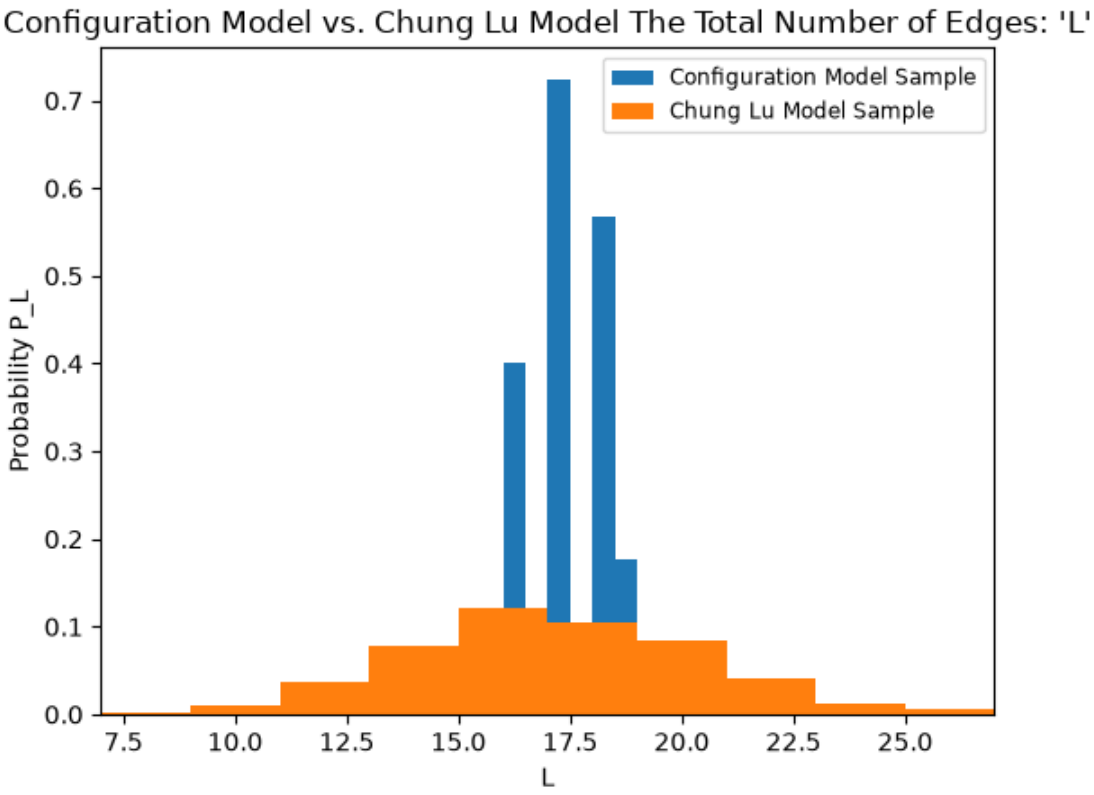
#---Plotting the Average Clustering Coefficient 'CC'
#Starting the plotting process
fig, ax = plt.subplots()#Note: fig is the plot container, while ax (i.e., the␣
 ↪'axes') is the plot itself
ax.hist(cc_list_config,label="Configuration Model␣
 ↪Sample",density=True)#Plotting the configuration model graph empirical data
ax.hist(cc_list_chunglu,label="Chung Lu Model Sample",density=True)#Plotting␣
 ↪the Chung Lu model graph empirical data
#ax.plot(L,P_l, color = 'r',label="Theoretical Distribution") #Plotting the␣
 ↪Theoretical
#Specifying figure details below:
handles, labels = ax.get_legend_handles_labels()
ax.legend(handles=handles, labels=labels,loc="upper right",fontsize="small")
ax.set_title("Configuration Model vs. Chung Lu Model Average Clustering␣
 ↪Coefficient 'CC'")
ax.set_xlabel("Average Clustering Coefficient")
ax.set_ylabel("Probability P_CC")
true_min=min(min(cc_list_config), min(cc_list_chunglu))
true_max=max(max(cc_list_config), max(cc_list_chunglu))
ax.set_xlim(true_min,true_max)
plt.show()
```

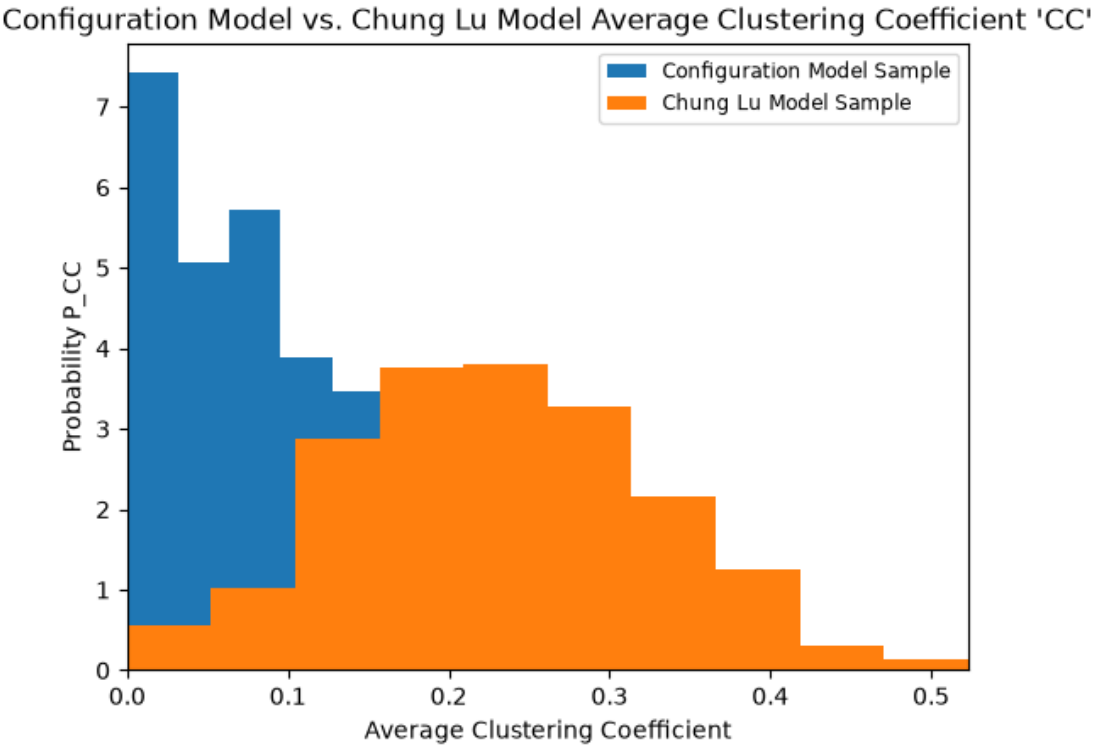
```
[(0, 5), (1, 3), (2, 3), (3, 4), (4, 1), (5, 2), (6, 2), (7, 4), (8, 2), (9, 0),
(10, 0), (11, 0), (12, 0), (13, 0), (14, 0)]
```

Depicting the configuration nodel graphically to show any self-loops that can occur:


10
11
8
5
14
6
0
2
1
7
3
12
9
4
13


Configuration Model
Chung Lu Model
0
2
4
6
8
10
12
14
0.0
2.5
5.0
7.5
10.0
12.5
1.0
0.8
0.6
0.4
0.2
0.0


Configuration Model vs. Chung Lu Model The Total Number of Edges: 'L'
Configuration Model Sample
Chung Lu Model Sample
Probability P_L
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
7.5
10.0
12.5
15.0
17.5
20.0
22.5
25.0
L


Configuration Model vs. Chung Lu Model Average Clustering Coefficient 'CC'
Configuration Model Sample
Chung Lu Model Sample
Probability P_CC
0
1
2
3
4
5
6
7
0.0
0.1
0.2
0.3
0.4
0.5
Average Clustering Coefficient

Discussing the Results: My heat map comparing the two models shows the general adjacency matrix value for each pair of nodes. Notably we see the Chung Lu model guaranteed no self-loops, while the Configuration Model had some. The general trend of the higher degree nodes (i.e., in this case node 0,1,2, etc., as I enforced and provided a sorted list from highest degree to lowest as input) remained the region with the highest overall adjacency matrix average values amongst the 1000 iterations. This indicates they had a high probability to be connected amongst one another, and in turn were thus often connected to one another. This is expected as the high degree nodes are the most likely for any random node to be connected to. In the case of Chung Lu however some of these pairs appeared to be connected close to if not 100% of the time, in contrast to the Configuration where these high-degree to high-degree pairs (e.g., node 0 to node 1), were the highest but did not occur 100% of the time across iterations.

As for my two metrics I chose for part c, I did the total number of edges 'L', and the average clustering coefficient. Here we can see the total number of edges, is approximately the same across graphs generated by both models, with both models coalescing around the same average total number of edges, of roughly ~17 or so, albeit at different probability levels. The average clustering coefficient on the other hand is much closer to 0 in the case of the configuration model while the Chung Lu model is relatively higher than 0. This is most likely due to the presence of self-loops and multi-edges that can occur in the configuration model. This artifact of the self-loops and multi-edges results in less clustering amongst nodes, relative to the Chung Lu model's differring approach.

**Problem 2 (20 points)**
In this question we will gain experience with community detection.

(a) Finding communities (5 points).

Download a network dataset from https://networkrepository.com/ with between roughly 1,000 $<$ N $<$ 5,000 nodes (select a different dataset from the one you used in assignment 1). Select a modularity maximization algorithm to detect communities in this graph.

(i) Visualize your network, coloring the nodes by community membership.
(ii) Report the modularity of the partition your algorithm found.
(iii) Report the number of communities detected.
(iv) Store your partition as a dictionary in the form of {node id: community id}.

(b) Null models (5 points).

Create a degree-preserving randomization of your network, and run the community detection algorithm you chose in part a on your randomized graph. Store each partition as a dictionary in the form of {node id: community id}.

(c) Plotting a null model comparison (5 points).

You should now have two dictionaries, corresponding to the partitions from parts a and b. Create a figure with two subplots: In each, plot the community size distribution of each partition. Your figure should be publication-quality, with well-labeled, consistent axes, log-scaled wherever relevant, etc.

(d) Analysis (5 points).

For the different partitions computed above, discuss how you would approach comparing this functions quantitatively. For example, imagine a function f(p1,p2) that takes partitions p1 and p2 as input and returns a number corresponding to how similar or dissimilar the partitions

are. What properties should this function f have? For this question, base your answer on a technique used in the literature, after a brief search about methods for quantifying differences between graph partitions. You do not need to implement the function, a high-level discussion is sufficient.

[18]:
```
from scipy.io import mmread

a = mmread('ca-CSphd.mtx')
phd_collabs =nx.DiGraph(a).to_undirected()
```

```
/tmp/ipykernel_156064/3807146788.py:3: DeprecationWarning: The default value for
`spmatrix` is changing to `False` in v1.20.
            That means the default return type will be a sparse array.
            Unless you use * instead of @, ** for matrix power, or you depend
            on 2D shapes from e.g. `A.sum(axis=0)` it may not matter to you.
            See the spmatrix to sparray migration guide for details.
https://docs.scipy.org/doc/scipy/reference/sparse.migration_to_sparray.html

  a = mmread('ca-CSphd.mtx')
```

[11]:
```
import networkx as nx
import numpy as np
from collections import defaultdict
from random import shuffle
import matplotlib.pyplot as plt
import community

pos = nx.fruchterman_reingold_layout(phd_collabs)
partition = community.best_partition(phd_collabs)

# Output the best modularity score
print("Best Modularity: ", community.modularity(partition, phd_collabs))

# Output the Number of Communities Detected {node_id: community_id}
print("Number of Communities Detected: ",len(np.unique(list(partition.
 ↪values()))))

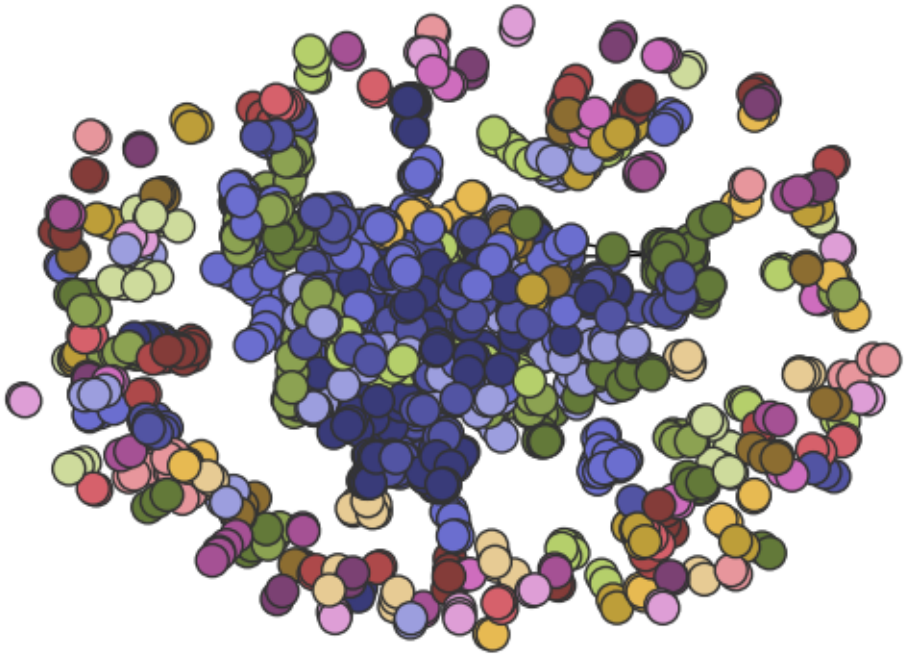
# Storing and showing partition as dictionary
phd_collabs_parts = dict(partition)
print("Partition Dictionary: ", phd_collabs_parts)
colors_sns = plt.cm.tab20b(np.linspace(0,1,len(np.unique(list(partition.
 ↪values())))))
colors_nodes = [colors_sns[partition[node]] for node in phd_collabs.nodes()]

fig, ax = plt.subplots(1,1,figsize=(7,5),dpi=100)
```

```
nx.draw(phd_collabs, pos=pos, node_size=150, edgecolors='.2',␣
 ↪node_color=colors_nodes, ax=ax)

plt.show()
```

```
Best Modularity:  0.9608196261068275 0.9608196261064889
Number of Communities Detected:  189
Partition Dictionary:  {0: 19, 1: 19, 2: 1, 3: 1, 4: 2, 5: 2, 6: 3, 7: 3, 8: 4,
}
```



[41]:
```
#(b)
degree_dict = dict(phd_collabs.degree())
degree_sequence = list(degree_dict.values())
null_phd_collabs=chung_lu_model_from_degree_sequence(degree_sequence)
pos = nx.fruchterman_reingold_layout(null_phd_collabs)
partition = community.best_partition(null_phd_collabs)
# Output the best modularity score
print("Best Modularity: ",community.modularity(partition, null_phd_collabs))

# Output the Number of Communities Detected {node_id: community_id}
print("Number of Communities Detected: ",len(np.unique(list(partition.
 ↪values()))))
```

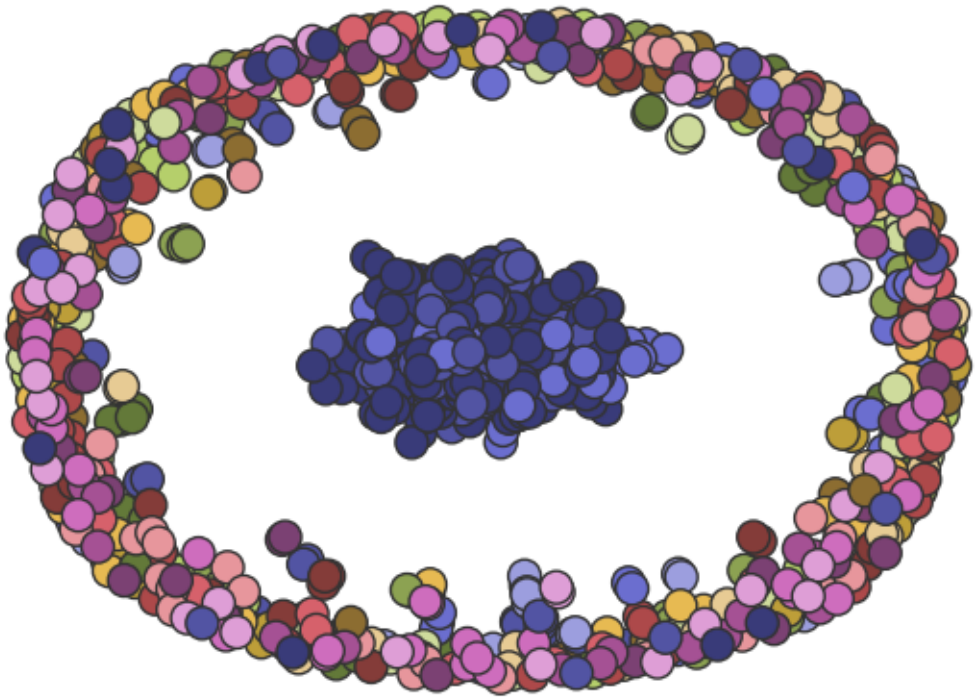
```
# Storing and showing partition as dictionary
null_phd_collabs_parts = dict(partition)
print("Partition Dictionary: ", null_phd_collabs_parts)
colors_sns = plt.cm.tab20b(np.linspace(0,1,len(np.unique(list(partition.
 ↪values())))))
colors_nodes = [colors_sns[partition[node]] for node in null_phd_collabs.
 ↪nodes()]

fig, ax = plt.subplots(1,1,figsize=(7,5),dpi=100)

nx.draw(null_phd_collabs, pos=pos, node_size=150, edgecolors='.2',␣
 ↪node_color=colors_nodes, ax=ax)

plt.show()
```

```
Best Modularity:  0.7232526057391191
Number of Communities Detected:  618
Partition Dictionary:  {0: 0, 1: 38, 2: 17, 3: 3, 4: 30, 5: 26, 6: 17, 7: 7, 8:
}
```



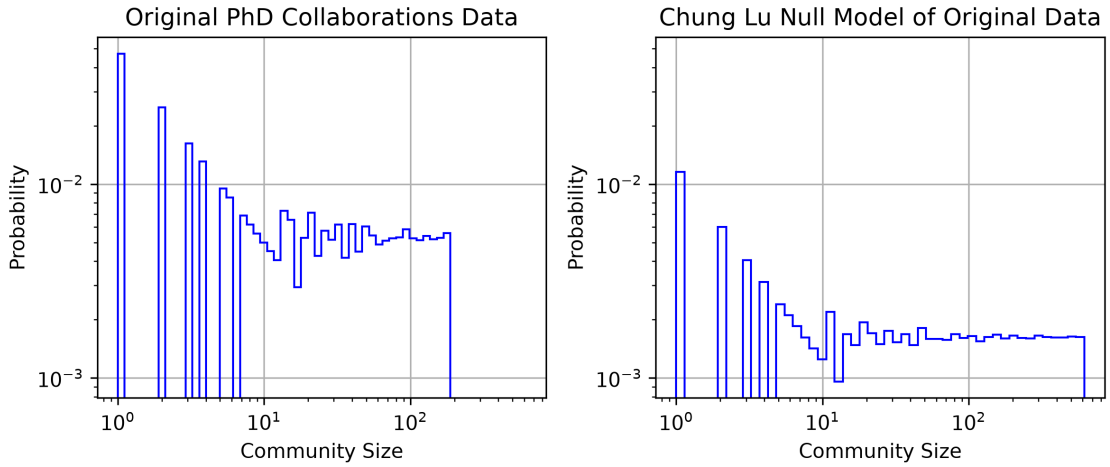
[51]:
```
from collections import Counter

phd_collabs_community_sizes = Counter(phd_collabs_parts.values())
null_phd_collabs_community_sizes = Counter(null_phd_collabs_parts.values())

fig, ax = plt.subplots(1,2,figsize=(8,3.5),dpi=300,sharey=True,sharex=True)

# Plotting the community size distribution of the original dataset
org_bins = np.geomspace(1,np.max(list(phd_collabs_community_sizes)))#Defining␣
 ↪the smallest community as an isolated node of 1
ax[0].hist(list(phd_collabs_community_sizes),bins=org_bins, density=True,color␣
 ↪= 'b',histtype='step')
ax[0].loglog()
ax[0].set_title("Original PhD Collaborations Data")
ax[0].set_xlabel("Community Size")
ax[0].set_ylabel("Probability")
ax[0].grid(True)
# Plotting the community size distribution of the Chung Lu Null Model of the␣
 ↪dataset
null_bins = np.geomspace(1,np.max(list(null_phd_collabs_community_sizes)))
ax[1].
 ↪hist(list(null_phd_collabs_community_sizes),bins=null_bins,density=True,color␣
 ↪= 'b',histtype='step')
ax[1].loglog()
ax[1].set_title("Chung Lu Null Model of Original Data")
ax[1].set_xlabel("Community Size")
ax[1].set_ylabel("Probability")
ax[1].tick_params(labelleft=True)
ax[1].grid(True)

plt.tight_layout()
```



Analysis:

In the literature there are a vast number of different approaches to comparing two partitions: (Fortunato, Santo. "Community Detection in Graphs." Physics Reports 486, no. 3 (2010): 75–174. https://doi.org/10.1016/j.physrep.2009.11.002.) provides a really nice overview of many!

I would likely approach the problem through the classification error H(X,Y) measure (M. Meilă, D. Heckerman, An experimental comparison of model-based clustering methods, Mach. Learn. 42 (1) (2001) 9–29.), which returns back the maximum overlap between the two partitions, in terms of community assignments of each node. This method seemed the most intuitive to me. Since, my intial analysis simply assigns a single community partition per node, this would allow me to compare directly to see how many nodes were assigned to the same community across the two partitions. This is good because my dataset reflects PhD Collaborations, and so being able to distinguish between a person who may be part of 1 community as opposed to definitely apart of 1 community is incredibly valuable in better understanding my dataset.

The only caveat I think would be that in real-life people tend to have multiple different communities, each of which they could be involved with at varying levels, and so by only evaluating the match across a single community assignment I miss some of that information, however to analyze assignments to multiple communities requires leveraging a different community assignment algorithm to the Louvain algorithm I utilized in my analyses above.

Overall, I think an ideal function, f, to approach this problem would likely extend the idea of classification error but to a list of all communities a given node was assigned to across the 2 different partitions, each with an associated probability score. This would provide a more robust measure of community assignments per node that would allow one to see the following properties: whether a node is generally a part of some subset of communities, or not, to what degree if so, and how confident each partition was in assigning said communities. I think these are the primary properties the function f should have in order to determine well-enough whether a given pair of partitions are similar or not.

**Problem 3 (20 points)**
In this question we will revisit degree sequences with a focus on preferential attachment. We have worked with G(n,p) random graphs extensively. We have also discussed how most graphs are not random, but follow some kind of preference for homophily or highly "desirable" nodes. One common preference is to connect to high degree nodes, reinforcing a rich-get-richer dynamic. We will create a network growth model, where we add new vertices to a graph, and create edges to existing vertices with a probability based on vertex i's degree ki: p(i) = k i + 1 n j=0(k j + 1) The parameter allows us to configure the model between a sublinear preference for high-degree nodes (0 < < 1), a linear preference ( = 1), and a superlinear preference ( > 1). The +1 in the above equation ensures some chance of connecting to any vertex, even if it initially has a degree of zero.

(i) Implement the generator (4 points).
Write a graph generating function preferentialGenerator(n, a) that returns a graph with n vertices and = a.

(ii) Degree analysis (4 points).
Generate graphs with 1,000 nodes and alpha values of 0.1, 1, and 2. Make a plot comparing the

three degree distributions. You may use histograms, kernel-density estimates, or cumulative distribution functions. However, your plots should be easy to interpret: select number of bins or bin widths, log scale, or other attributes as needed to produce a high-quality figure. We will now pivot to the random walk model of network generation. In this model, every time we add a vertex i we add two edges. First, an edge to an existing random vertex j. Then, with probability p, we add an edge from i to a random neighbor of j. Otherwise we instead add an edge to another random vertex in the graph.

(iii) Implement the generator (4 points).

Write a graph generating function randomWalkGenerator(n, p) that returns a graph with n vertices generated by random walks with probability p.

(iv) Clustering analysis (4 points).

Generate graphs with 1,000 nodes and p values of 0, 0.1, 0.5, and 0.9. Make a plot comparing the four clustering distributions. As in part (ii), make these plots easily interpretable. And finally…

(v) Interpretation (4 points).

Compare the graphs created by the two network growth models. What kind of structures do they produce? In what circumstances might we want to use preferentialGenerator as a null model, and when might the random walk model be a more appropriate null assumption?

[ ]:
```
def network_growth_model(N,alpha):
    """
    Generates a Network Growth Model using preferential attachment.

    Parameters
    ----------
    N : int
        The number of vertices of the graph to generate.

    alpha : float
        The average number of edges to make for each node added to the graph

    Returns
    -------
    G : networkx.Graph
        A graph generated from the given number of nodes and connections per␣
 ↪node added (alpha).


    Notes
    -----
    -
    Time Complexity
    ---------------
    The time complexity is dominated by O(N**2), where N is the number of nodes␣
 ↪we want to add to the graph,
    each with an 'alpha' number of connections. Where alpha is bounded by N the␣
 ↪number of new nodes we
```

```
    want to provide alpha connections for.

    Example
    -------
    >>> N = 100
    >>> alpha = 3
    >>> G = network_growth_model(N,alpha)
    """
    #Instantiate G
    G = nx.Graph()
    #Instantiate at least 1 node
    G.add_nodes_from(range(1))

    #Concept:
    #For every new node we want to add,
        #First if there are no connections then just add connect to what we initialized with
    #Find the probabilities of connecting to all the existing nodes currently
    #Pick 'alpha' nodes at random based on said probabilities
    #Add a new node and connect it to each of the list of 'alpha' nodes
    #Repeat for all 'N' new nodes

    while N>0:#While we still have nodes to add...
        nodes = list(G.nodes())#Get the current list of nodes...
        degs = np.array([(G.degree[n]) for n in nodes], dtype=float)#And, get their probabilities.

        #Translating alpha into an N value so that on average 'alpha' connections are made even if this is a float
        m = int(np.floor(alpha)) + (np.random.rand() < alpha - np.floor(alpha))

        #Edge Case: If this is the first run where we start with a Graph with no connections
        if degs.sum() == 0:
            chosen = nodes #then just instantiate to the few nodes we initialized with
        else:
            #Otherwise, pick 'm' connections to make **without replacement**,
            #  by the probability calculated from their degree(s)
            chosen = np.random.choice(nodes, size=m, replace=False, p=degs / degs.sum())
        #Add the new node as the next id number
        new_id = len(nodes)
        #And connect it to 'alpha' chosen nodes
        for node_i in chosen:
            G.add_edge(int(node_i), new_id)
```

```
        #Update N to reflect that we processed 1 new node!
        N-=1

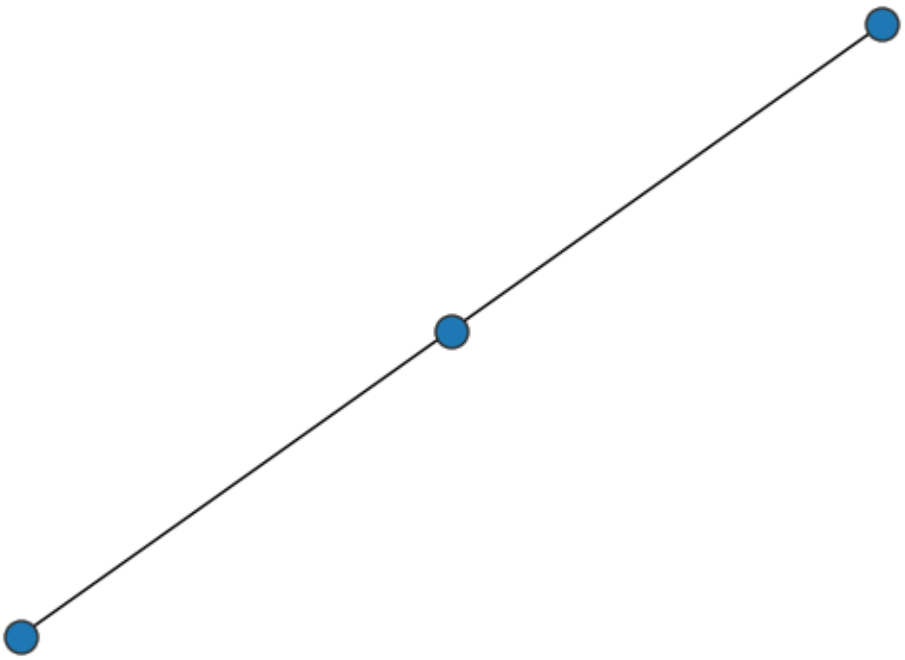
    return G
```

[14]:
```
#Testing
import matplotlib.animation as animation
N = 10
alpha = .1
G = network_growth_model(N,alpha)

print(G.degree())
fig, ax = plt.subplots(1,1,figsize=(7,5),dpi=100)

visual = nx.draw(G, pos=nx.spring_layout(G), node_size=150, edgecolors='.2',␣
 ↪ax=ax)
```

```
[(0, 1), (1, 2), (2, 1)]
```



[29]:
```
N = 1000
alpha1 = .1
alpha2 = 1
alpha3 = 2
```

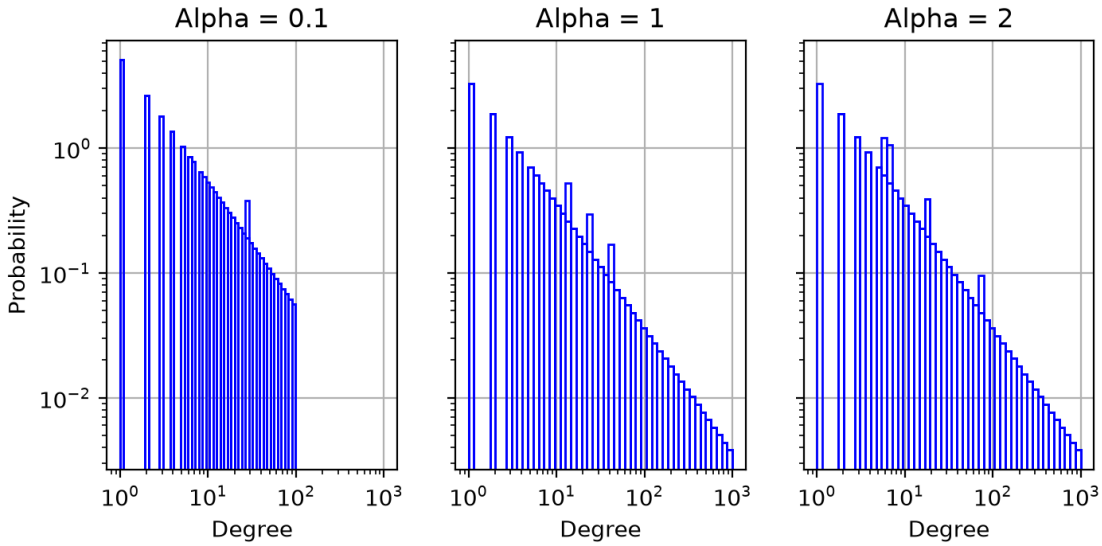
```
G1 = (network_growth_model(N,alpha1))
deg1 = list(G1.degree())
G2 = (network_growth_model(N,alpha2))
deg2 = list(G2.degree())
G3 = (network_growth_model(N,alpha3))
deg3 = list(G3.degree())
```

[37]:
```
fig, ax = plt.subplots(1,3,figsize=(8,3.5),dpi=200,sharex=True,sharey=True)

deg_bins1 = np.geomspace(np.max([np.min(deg1),1]),np.max(deg1))
ax[0].hist(deg1,bins=deg_bins1, density=True,color = 'b',histtype='step')
ax[0].loglog()
ax[0].set_title("Alpha = 0.1")
ax[0].set_xlabel("Degree")
ax[0].set_ylabel("Probability")
ax[0].grid(True)

deg_bins2 = np.geomspace(np.max([np.min(deg2),1]),np.max(deg2))
ax[1].hist(deg2,bins=deg_bins2, density=True,color = 'b',histtype='step')
ax[1].loglog()
ax[1].set_title("Alpha = 1")
ax[1].set_xlabel("Degree")
#ax[1].set_ylabel("Probability")
ax[1].grid(True)

deg_bins3 = np.geomspace(np.max([np.min(deg3),1]),np.max(deg3))
ax[2].hist(deg3,bins=deg_bins3, density=True,color = 'b',histtype='step')
ax[2].loglog()
ax[2].set_title("Alpha = 2")
ax[2].set_xlabel("Degree")
#ax[2].set_ylabel("Probability")
ax[2].grid(True)
```




[ ]:
```
def random_walk_generator(N,p):
    """
    Generates a Random Walk Model

    Parameters
    ----------
    N : int
        The number of vertices of the graph to generate.

    p : float
        The probability we make a second edge of each new vertex, a neighbor of␣
↪the vertex first connected to.

    Returns
    -------
    G : networkx.Graph
        A graph generated from the given number of nodes and connections per␣
↪node added (alpha).


    Notes
    -----
    -
    Time Complexity
    ---------------
    The time complexity is dominated by O(N), where N is the number of nodes we␣
↪want to add to the graph,
    each with 2 connections.
```

```
    Example
    -------
    >>> N = 100
    >>> p = .33
    >>> G = random_walk_generator(N,p)

    """
    #Instantiate G
    G = nx.Graph()
    #Instantiate at least 2 nodes (that are connected)
    G.add_edge(0,1)

    #Concept:
    # For every vertex we want to add (out of N),
    #  we add 2 edges. (i.e., m=2)
    # First edge is chosen at random (assuming no self loops?) n choose 1
    # Second edge is, if probability p, get random neighbor of j (i.e., n␣
↪choose k_j)
    # else,
    # just repeat first edge
    #

    while N>0:#While we still have nodes to add...
        nodes = list(G.nodes())#Get the current list of nodes...
        new_node= len(nodes)
        vertex_j = np.random.choice(nodes, size=1)
        nodes.remove(*vertex_j)#Prevent duplicate edges
        if np.random.rand() > p:
            G.add_edge(int(*vertex_j),new_node)
            vertex_j = np.random.choice(nodes, size=1)
            G.add_edge(int(*vertex_j),new_node)
        else:
            vertex_j_neighbor = np.random.choice(list(G.neighbors(*vertex_j)),␣
↪size=1)
            G.add_edge(int(*vertex_j),new_node)
            G.add_edge(*vertex_j_neighbor,new_node)
        N-=1
    return G
```

[175]:
```
#Testing
import matplotlib.animation as animation
N = 1
p = 0.3
G = random_walk_generator(N,p)

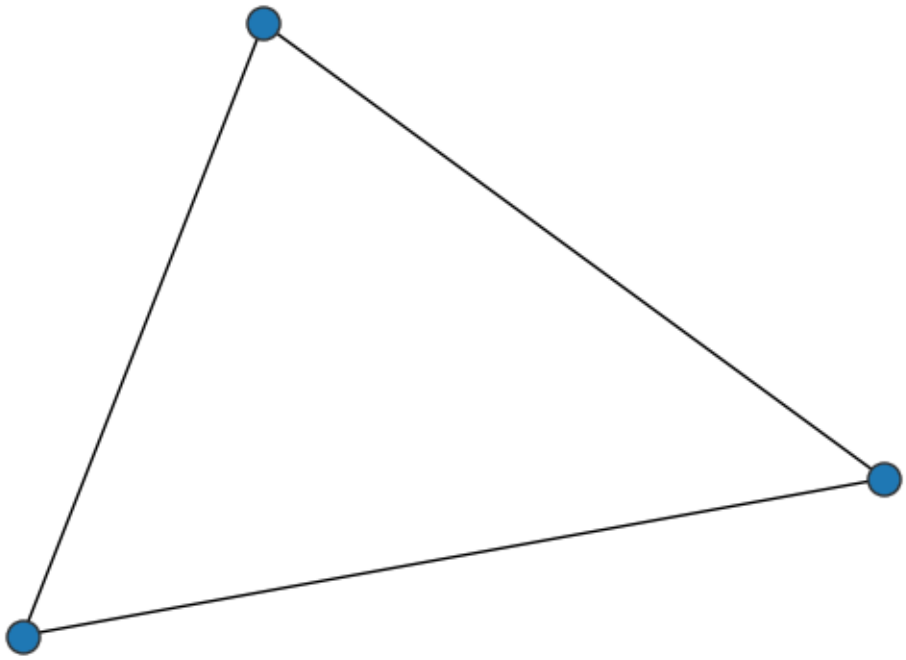
# Expectations:
```

```
# whenever we roll a chance < p, then we will be connected to an edge and its␣
 ↪neighbor. i.e., v_ij1 will have an edge with v_ij2, and n_i will have edges␣
 ↪with both of them
# We will have N+2 nodes (we need a minimum of two nodes to start connecting 2␣
 ↪edges for every vertex)
# We will have a minimum of 2 edges per node. N=1 should be a perfect triangle

print(G.degree())
fig, ax = plt.subplots(1,1,figsize=(7,5),dpi=100)

visual = nx.draw(G, pos=nx.spring_layout(G), node_size=150, edgecolors='.2',␣
 ↪ax=ax)
```

```
[(0, 2), (1, 2), (2, 2)]
```



[176]:
```
N = 1000
p1 = 0
p2 = 0.1
p3 = 0.5
p4 = 0.9

G1 = (random_walk_generator(N,p1))
```

```
cc1 = list(dict(nx.clustering(G1)).values())
G2 = (random_walk_generator(N,p2))
cc2 = list(dict(nx.clustering(G2)).values())
G3 = (random_walk_generator(N,p3))
cc3 = list(dict(nx.clustering(G3)).values())
G4 = (random_walk_generator(N,p4))
cc4 = list(dict(nx.clustering(G4)).values())
```

[ ]:
```
fig, ax = plt.subplots(1,4,figsize=(8,3.5),dpi=200,sharex=True,sharey=True)

#Since there are cases where the minimum CC is 0,
# I'm starting the plot with the second-most minimum

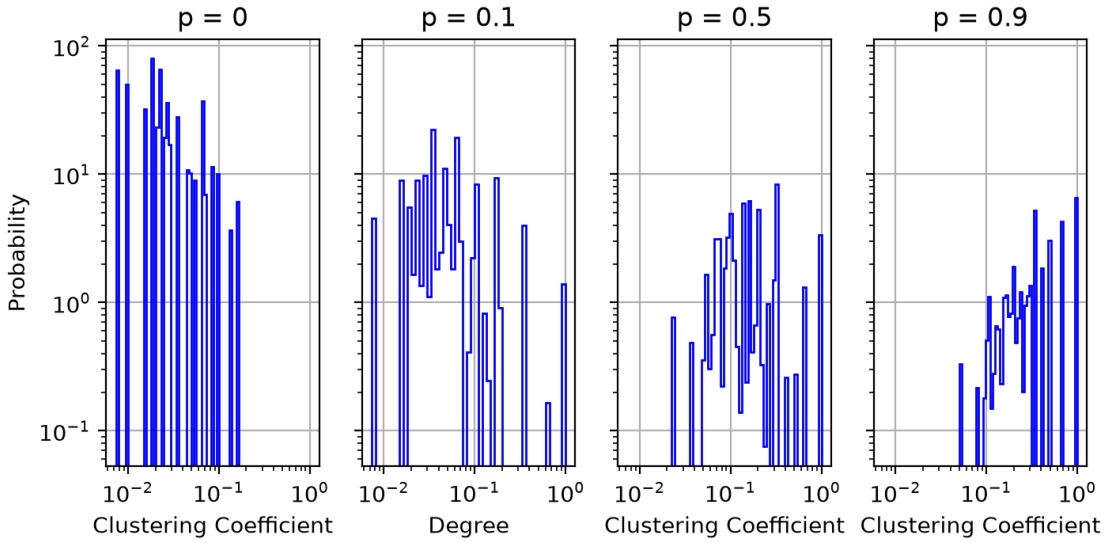
#Gathering 2nd most mins
cc1=np.asarray(cc1)
cc1_min2= np.amin(cc1[cc1 != np.amin(cc1)])
cc2=np.asarray(cc2)
cc2_min2= np.amin(cc2[cc2 != np.amin(cc2)])
cc3=np.asarray(cc3)
cc3_min2= np.amin(cc3[cc3 != np.amin(cc3)])
cc4=np.asarray(cc4)
cc4_min2= np.amin(cc4[cc4 != np.amin(cc4)])

cc_bins1 = np.geomspace(cc1_min2,np.max(cc1))
ax[0].hist(cc1,bins=cc_bins1, density=True,color = 'b',histtype='step')
ax[0].loglog()
ax[0].set_title("p = 0")
ax[0].set_xlabel("Clustering Coefficient")
ax[0].set_ylabel("Probability")
ax[0].grid(True)

cc_bins2 = np.geomspace(cc2_min2,np.max(cc2))
ax[1].hist(cc2,bins=cc_bins2, density=True,color = 'b',histtype='step')
ax[1].loglog()
ax[1].set_title("p = 0.1")
ax[1].set_xlabel("Clustering Coefficient")
#ax[1].set_ylabel("Probability")
ax[1].grid(True)

cc_bins3 = np.geomspace(cc3_min2,np.max(cc3))
ax[2].hist(cc3,bins=cc_bins3, density=True,color = 'b',histtype='step')
ax[2].loglog()
ax[2].set_title("p = 0.5")
ax[2].set_xlabel("Clustering Coefficient")
#ax[2].set_ylabel("Probability")
ax[2].grid(True)
```

```
cc_bins4 = np.geomspace(cc4_min2,np.max(cc4))
ax[3].hist(cc4,bins=cc_bins4, density=True,color = 'b',histtype='step')
ax[3].loglog()
ax[3].set_title("p = 0.9")
ax[3].set_xlabel("Clustering Coefficient")
#ax[3].set_ylabel("Probability")
ax[3].grid(True)
```




Interpretation: The network-growth model produced degree distributions that follow a power law curve (i.e., a linear shape on a log-log plot). This means it produced networks structures where hubs emerged and tended to be the places where new nodes made connections to. The randomWalkGenerator on the other hand allowed us to generate various levels of clustering coefficient distributions by picking different p values. A preferentialGenerator may be a better suited null model therefore when we want to simulate or model a network where we know there are strong hubs that dominate the network. Take the internet for example, it's a well-known fact that the most popular websites dominate most searches and webpage requests online, and as such many new websites may promote their own website through ads on this site (i.e., seek out a connection to it) through for example, running ads on Google to garner more attention, or, posting an opinion piece on a well known newpaper website to bring attention to your own, etc. Conversely, a randomWalkGenerator may be more helpful when you are looking at perhaps a relatively smaller but known to be tighter knit network. Say, a circle of friends, people, researchers, etc. Here one can leverage the randomWalkGenerator by tuning the p-value to the level of clustering observed in the groups of interest, to observe this level of clustering amongst a network of these tight-knit circles

**Problem 4 (20 points)**
We are training you to conduct original research, and a crucial component of that is a literature review. We can conduct literature reviews for several reasons, including familiarizing ourselves

with a topic of study, a set of related methods, or the conventional wisdom of a field. When beginning a research project we typically want to know whether our research question has been asked before (and if so, using what tools and with what data), what related questions have been asked, and what gaps in understanding remain that we can try to answer. Today we will start small. Search Google Scholar for five papers on different community detection algorithms. Don't use the ones we've discussed extensively in class (label propagation, Louvain, Girvan-Newman). For each algorithm, answer the following:

(a) What measure is being optimized (how do we know a given partition is optimal or good? Are they use modularity maximization, or something else?)
(b) Algorithmically (a high level explanation in a few sentences is fine), how are partitions selected / sampled?
(c) When did this technique come out? Does the journal or conference suggest what field this al- gorithm is used in? Is it terribly obscure? (We can use citation count as a proxy for obscurity here)
(d) Is the algorithm particularly well-suited to certain kinds of graphs? (Does it assume sparse graphs? A core-periphery structure? Does it scale well / poorly? Add a little context about where you might / should not use this algorithm)

I encourage you to read selectively; you want to understand how these algorithms work and where and how they could be used, but you don't need to understand every check for division by zero or the full literature review of how this approach has been used in MRI analysis and how detected neuronal communities differ from established medical literature. Read papers for what you need from them.

5 Papers on Community Detection:

1. "Detection of Overlapping Communities in Dynamical Social Networks | IEEE Conference Publication | IEEE Xplore." Accessed October 30, 2026. https://ieeexplore.ieee.org/abstract/document/5591234.

(a) There are 2 values that are used to characterize communities (i.e., optimized for), the mean number of second neighbors, which reflects the mean number of neighbors that the community is at least a path length or 2 or less from, and the mean number of robust second neighbors, which reflects the mean number of neighbors that the community has at least 2 paths of length or 2 or less from. (i.e., that there is more than 1 path that places it on the former list).

This is meant to capture: * Ability to access easily (in two steps) to most of the community (at least as much as the others nodes of the community) * Robust access to other nodes of the community (at least as much as the others nodes of the community).

These values are optimized for adpatively to ensure the same community can be detected whether the original whole graph is relatively sparse or conversely relatively dense. The authors argue other algorithms tend to produce differing results because the density of the whole graph is often used to create what they call a "relative" definition of community.

(b) For every time-stamped set of edges, and each edge within said set, attempt to integrate (i.e., compare against the 2 index values) any nodes outside the community (i.e., taken from the edge) into the community. Once complete, update previously made communities. Then, If neither node (off the edge) is a part of a community create a new community. Lastly, merge similar communities. (i.e., check the percentage of nodes that overlap, between a pair of

communities, against a definiable threshold)

(c) This algorithm came out in 2010 in an IEEE conference. It is intended specifically to address the problem of being able to capture overlapping communities which can happen in multiple fields, but here they key-in on social networks in particular. It has 218 citations so it's not terribly obscure.

(d) This algorithm is particularly well-suited to ensure results that are agnostic of the graph density. It allows you to define a minimial community as a kind of pre-defined pattern. In other-words defining the smallest community you'd like to detect!

2. Aghaalizadeh, Saeid, Saeid Taghavi Afshord, Asgarali Bouyer, and Babak Anari. "A Three-Stage Algorithm for Local Community Detection Based on the High Node Importance Ranking in Social Networks." Physica A: Statistical Mechanics and Its Applications 563 (February 2021): 125420. https://doi.org/10.1016/j.physa.2020.125420.

(a) The main measure used in LCDR, is local community detection via high importance nodes ranking (RDC). The RDC index for the node indicates its local importance based on the degree of the neighboring nodes. This is calculated from thinking about important nodes as those that are producers of information, that can be measured via Kirchoff's law (i.e., analagous to currents through resistors in a circuit).

(b)

1. "Select a number of nodes as the important nodes based on the proposed ranking and initial community formation."
2. "Expand the initial communities by allocating the unselected nodes based on the proposed proximity ranking with important nodes."
3. "Merge initial communities based on the proposed relation and extract the final communities."

(c) This algorithm came out in 2020 through a Physica journal. It is intended to produce better stability, accuracy and performance and scalability than other local/fast algorthms. (i.e., O(n) time). It's somewhat obcure at 55 citations, though it is more recent then the previous paper

(d) The algorithm is intended for use in graphs when there is no prior knowledge beforehand and speed is desired. It can significantly improve accuracy relative to other local, and fast algorithms; when compared on simulated and real-world networks.

3.ACM Other Conferences. "Efficient Community Detection in Large Networks Using Content and Links | Proceedings of the 22nd International Conference on World Wide Web." Accessed October 30, 2026. https://dl.acm.org/doi/10.1145/2488388.2488483.

(a) This algorithm intended to address the problem that many networks can actually contain noise presented as link structure, and that by synthesizing link structure with content, a stronger community signal can be better produced. In this sense they use both an estimate on whether a given link is at high probability to reside in a given community, and that it's content is similar to other nodes in the community as determined via cosine similarity or Jaccard coefficient.

(b) The idea is to: "

1. Create content edges.
2. Sample the union of content edges and topological edges with bias, retaining only edges that are relevant in local neighborhoods.

3. Partition the simplified graph into clusters." Where content edges are edges that can be draw based on content similiarity where as topological edges are those drawn more conventionally (e.g., whether the two nodes had contact)

(c) This technique came out in 2013 as a part of a proceeding for the international conference on the world wide web. It has 248 citationsso it's not terribly obscure, and is intended specifically to address noise: i.e., false positives and negatives that are common in online social networks and information networks.

(d) It's particularly well-suited to real-world networks where the nodes reflect social media or webpages that hold inherent content or hyperlinks that can be used to compare one another for similarity.

4. Airoldi, Edoardo M., David M. Blei, Stephen E. Fienberg, and Eric P. Xing. "Mixed Membership Stochastic Blockmodels." arXiv:0705.4485. Preprint, arXiv, May 30, 2007. https://doi.org/10.48550/arXiv.0705.4485.

(a) This algorithm was made to address the inherent interdependent nature of relational data in real-world networks that runs counter to standard assumptions of, for example, communities being purely independent. It optimizes for a kind of membership probability vector, where each state of the vector represents a probability that said node may be part of each given community.

(b) The high-level overview is: for each node, first draw a mixed membership vector. Then for each pair of nodes, and for each outgoing edge draw a probability of membership for the holder of said outgoing edge (i.e., the initiator), and for each incoming edge draw a probability of membership for the holder of the incoming edge (i.e., the receiver). From these two values sample their interaction, as collapsed via Bernoulli's to determine if they share membership or not.

(c) This technique came out in 2007 in a NeurIPS Proceeding Advances in NIPS. It has 2959 citations making it quite popular!

(d) It's particularly well-suited for networks where membership in communities is known interdependent (i.e., interelated) and establishing multiple-communities per node is desired.

5. Zong, Xulin, Xiangqian Ding, and Zhen Zhou. "Health Level Classification by Fusing Medical Evaluation from Multiple Social Networks." Future Generation Computer Systems 114 (January 2021): 574–80. https://doi.org/10.1016/j.future.2020.08.008.

(a) This algorithm is intended to optimize for accurate prediction of the health level of a user. It works by characterizing a user across multiple soureces to assign a vector of associated attributes to the user. The attributes are assigned by extracting known kinds of medical features from a user's content (e.g., analyzing image content or leveraging NLP). Partitions are known to be good by synthesizing data across multiple similar network representations of which the user is a part of. E.g., multiple websites.

(b) Communities and partitions are assigned by constructing a multiple affinity graph to describe a vast set of potential users. Users determined to be similar across the various attributes to other users are classified as a part of a kind of "medically-aware" cluster that is intended to represent a kind of set of similar health-statuses.

(c) This technique came out in 2021 as a part of a journal in Future Generation Computer Systems. It has only 5 citations making it pretty obscure!

(d) It's best when there are multiple different related datasets that contain medical data which the algorithm can leverage to cross-reference against one another, and analyse to determined

health status via online content of individuals.

**Problem 5 OPTIONAL (5 bonus points)**
For every problem set this semester, there will be an opportunity to earn extra credit. Since this is a new(-ish) course, we are eager to receive genuine feedback on the material, pace, instruction style, etc. of the course so far. Please provide any feedback about the lectures in recent weeks and/or this assignment

I'm still really enjoying the course! I think the coding exercises here have been good for me to develop better software development practices especially, I've been getting in the habit now of writing out expected behavior of the code I write as a means to do better and more robust testing. I think as we've entered more advanced content relative to the beginning of the semester I've also been reading through the textbook more carefully and taking more time to understand the mathematical theory. As an engineering background the formulations aren't always my forte so I really appreciate the time we take to explain them and really understanding what's going on in each concept and command we tackle.

I also really appreciated the open-ended lecture we had recently to just provide a forum to give status updates on projects and get feedback! I think in future iterations of the class having another periodic update or two like that throughout the semester could be really beneficial! Especially early-on in the semester and perhaps before the end deliverable.

I think as far constructive feedback, I don't have too much to add relative to last assignment. Something I think might be nice, could be to call-out textbook chapters alongside the lecture-topics in the syllabus, as I think that could help me in being more prepared for content ahead of each lecture! Personally I like the structure of knowing, topic X is all contained in ChX, so that's something I've been doing more lately (i.e., looking up the associated chapter in the textbook) that I think has been really helpful for the assignments!

Overall, as I said before I'm really enjoying the course, I feel like I'm learning a ton, and I think just wish I had more hours in the day to experiment with it all hahaha

Hope this feedback is of help, and looking forward to the rest of the course!